

## Contribution Journal

Ranveer Chand

For this lab, I started by uploading the notebook to Google Colab and sharing it with all group members, so everyone had access to it. My part was the setup code cell, which included the imports, generating the synthetic dataset, and plotting it before we moved into the polynomial regression models. Our group discussed all the reflection questions together, and I was mainly focused on the second question, about the learning curve for the underfitting model. Once everyone had their parts done, I created all the required files, added their parts to the contribution word document, downloaded the files in the correct formats, made sure everything was named correctly, and uploaded everything to our GitHub repository for submission.

100% contribution

John Nasir

By running the polynomial regression models with degrees 1, 4, and 15, I was able to clearly see how model complexity affects the shape of the predictions. My code for task 3/1 let me compare all three fits on the same plot, which made the differences easy to understand. The degree-1 model showed underfitting because it couldn't follow the curved pattern in the data. The degree-15 model showed overfitting because it tried too hard to match every noisy point. The degree-4 model landed in the middle and gave a smooth curve that matched the real trend. Creating and plotting these models helped me understand why choosing the right degree matters and how visualizing the results can make underfitting and overfitting much easier to spot.

100% contribution

Trevor Oakum

In Lab 08, I did Task 2, Part 4, along with my classmate, Freddy. Freddy and I worked through this section, then checked each others work to make sure we were satisfied with our work. Once then, we collaborated with the rest of the group to see if our code worked as intended to the rest of the lab, and if our other group members were satisfied with it.

100% contribution

Freddy Rodriguez

For Part 4 Task 2, I made learning curves for the degree 1 (underfitting) and degree 15 (overfitting) models using the `learning_curve` function and 5-fold cross-validation. I found a bug where the graphs weren't showing up right because the y-axis wasn't set, so I fixed that by adding `ylim=(-1, 1.01)` to both plots. Then I looked at the graphs to explain how the underfitting model had low scores that just flattened out (high bias), and how the overfitting model had a big gap between the training score and validation score (high variance).

100% contribution